TRUEWAY

Offline AI Travel Companion

DEEP RESEARCH / PRODUCT BLUEPRINT / TECHNICAL FEASIBILITY

From a landmark photo to trusted history,
even when the network disappears.

Includes: product strategy, 2026 technology assessment, regional packs, AI architecture, legal/licensing review, Bihar pilot, cost scenarios and delivery roadmap.

10 OCTOBER 2026 | RESEARCH EDITION 1.0

An independent planning study, not a performance benchmark or legal opinion.

NAVIGATE THE REPORT

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RESEARCH SYNTHESIS

Executive recommendation

> **Decision: build a compact, citation-first Android MVP**
>
> TrueWay should recognize **supported landmarks** with a light on-device vision model and optional offline GPS, retrieve verified facts from a downloadable regional pack, and generate friendly explanations only when a compatible offline LLM is available. Do not promise identification of every arbitrary place or universal offline navigation.

The fastest credible entry point is a **Bodh Gaya - Rajgir - Nalanda heritage pilot** with 50-100 carefully documented points of interest, plus a second small benchmark region to test whether the approach generalizes. The app must offer manual search where photography is prohibited or the image model is uncertain. [18, 19, 29, 30]

Make now	Make after validation	Defer / avoid
Camera scan, location candidates, verified history	Audio tour, OCR, multilingual explanations	Global all-place identification promise
Region pack download, install/repair/delete	Optional 388 MB visual-text embedder evaluation	Mandatory multi-GB AI model at install
Abstention, citations, offline search, travel journal	On-device question answering and trails	Unverified AI-generated historical reconstructions

Five findings that change the product plan

- **Offline is achievable, but model size dictates inclusivity:** LiteRT-LM lists a 388 MB text-vision EmbeddingGemma 2 variant and a 2,583 MB Gemma 4 E2B chat model. These are optional-component candidates, not universal device requirements. [1,2,3]
- **Landmarks must be treated as a closed/open-set recognition problem:** use ranked candidates plus a “not sure” outcome. Google Landmarks v2 is a research benchmark; a curated pilot corpus is still needed. [4]
- **Existing apps already provide offline knowledge, maps and audio tours:** TrueWay wins only if image-to-place-to-trusted-history works smoothly with the radio switched off. [11-15]
- **Rights are part of the technical design:** OSM data is ODbL, Wikidata statements are CC0, UNESCO data is CC BY-SA, and individual photos may demand attribution or additional permission. [6-10]
- **The field setting changes UX:** Bihar Tourism explicitly lists mobile/camera restrictions at Mahabodhi Temple. Search/nearby mode is mandatory, not a fallback hidden in settings. [18]

Evidence key

Tag	Meaning
Verified	Supported by a linked primary-source page, identified by [reference number].
Design decision	This report recommends an approach; it has not been implemented or evaluated.
Planning assumption	Numerical engineering, schedule, and cost estimates must be validated by experiments and quotations.

Scope of this study: strategic and technical feasibility, not a completed prototype. No custom AI model was trained, no field trial conducted, and no pricing quotation collected. Source links and review dates are in the reference catalogue.

CHAPTER 01

Concept, requirements & boundaries

Translate the handwritten vision into measurable product constraints.

Problem definition

When visiting a historical site, travellers often lack reliable connectivity, suitable guide availability, or prior context. A photograph can be an intuitive entry point, but a photograph alone is not a guarantee of identity. Offline knowledge must therefore be paired with location context and conservative decision rules.

Jobs to be done

- **Identify:** “What am I looking at?” - return a probable landmark with a confidence explanation.
- **Understand:** “Why is it important?” - give a short, sourced answer, then expandable architecture and history.
- **Explore:** “What else should I see nearby?” - provide an offline list of POIs and a basic map.
- **Remember:** “What did I discover?” - preserve photographs and visited places privately on device.
- **Respect:** “What are the local rules?” - surface cultural etiquette and photography restrictions.

MVP invariants

Requirement	Acceptance criterion
True offline mode	After completed installation, camera match, local search, fact display, and journal work in airplane mode.
Know when unknown	Out-of-region or ambiguous images lead to “No reliable match” or candidate selection, not invented history.
Attribution	Every substantial historical claim has at least one locally stored source ID and source metadata.
Small base install	The first useful mode does not depend on downloading a 2.5 GB model.
User-controlled storage	Pack sizes are disclosed, updates are resumable, and deletion is supported.
Site compliance	Manual lookup is available whenever capturing an image is disallowed.

Explicit non-goals

The MVP does **not** promise global coverage, accurate identification of anonymous buildings, instant translation of ancient inscriptions, live opening hours, precise indoor positioning, or emergency-grade navigation. Those problems need separate datasets and testing.

CHAPTER 02

Market signals and competitive landscape

Validated demand signals are not the same as a proven TrueWay market.

Demand indicators - use with care

India's Ministry of Tourism dashboard reports **4,286.9 million domestic tourist visits in 2025** and about **1,534 million international tourist arrivals worldwide in 2025**. These are trip/visit metrics, not unique TrueWay customers or evidence of willingness to pay. The official dashboard is labelled beta. [22]

An earlier official 2025 tourism compendium reported **2,948.19 million domestic tourist visits in India in 2024**; report revisions and definitions should be tracked when preparing a pitch. [23]

Competitive reality

Option	Established job	TrueWay gap to own
Google Maps	Download regions for offline map use; no offline transit, cycling or walking directions in its standard flow. [11]	Camera-led heritage retrieval + sourced narratives.
Google Lens	Search by image, with search and similar-image results. [15]	Deliberately preloaded, self-contained recognition + knowledge.
Kiwix	Download compressed ZIM knowledge content for offline reading. [12]	Instant context: "what is in front of me?"
izi.TRAVEL	City/museum audio tours; downloadable tours and GPS triggers. [13]	On-demand photo recognition beyond prepared tour stops.
VoiceMap	Downloaded route, audio and maps with GPS-triggered narration. [14]	Flexible exploration without depending on a fixed route.

Positioning: "The offline visual heritage guide with sources you can inspect." Avoid framing as the world's first offline guide. The benchmark competitor is the traveller's existing combination of Maps + camera search + downloaded article/audio, not only standalone AI apps.

Target personas and priority

Persona	Situation	Core need
Independent heritage traveller	Visits monuments and walks outside urban networks.	Fast explanation, offline map, reliable identity.
Architecture / history student	Studies forms, materials, historical phases.	Evidence-backed architectural elements and references.
International pilgrim	Travels sacred sites with mixed connectivity.	Multilingual narration, etiquette, predownloaded content.
Local guide / museum	Wants a reusable curated guide.	Verified regional packs and content-authoring tools.

Sustainable advantage

A defensible advantage is not an LLM wrapper. It is a **licensed, versioned local heritage graph + robust open-set recognition + field-tested travel UX**. Competitors can copy features; better curatorial coverage, provenance, offline performance and partnerships are harder to reproduce.

CHAPTER 03

Feature strategy and release scope

Plan by user value, implementation risk and offline dependency.

P0: launch	P1: expand	P2: differentiate
Photograph landmark and rank candidates	Offline AI question-answering grounded in pack	AR historical reconstruction with provenance
Manual title/category search	Architecture detective with image region explanations	Audio/text multimodal local AI
History summary + verified sources	OCR of boards; predownloaded translation	Peer transfer with signed-pack verification
GPS-aware nearby landmarks	Curated audio narration, multilingual	Crowd contributions with human review
Download, verify, update and delete packs	Offline itinerary / walking trail	Indoor beacon positioning / museum mode
Journal and bookmarking	Cultural etiquette and accessibility extensions	Cross-country pack marketplace

Feature-to-architecture traceability

Feature	Dependencies	Failure-safe behavior
Photo identification	Model, reference embeddings, optional GNSS	Ranked alternatives; manual search.
History view	Place ID, source IDs, licensed text	Display prewritten text without LLM.
Offline nearby	Local spatial index, GNSS permissions	Browse region map if GPS unavailable.
Ask an AI	Downloaded chat model, retrieval, facts	Hide AI button or use FAQ when model absent.
OCR and translate	OCR model, language pack	Retain original text and warning on limitations.
Offline maps	Permitted map tiles/geometry	Text place list still operates.

Prioritization rule

Choose features by (i) offline completion rate, (ii) user value at a heritage site, (iii) evidence and licensing readiness, (iv) device compatibility, and (v) development burden. The wow-factor mode should never compromise the simple “scan / identify / read” path.

CHAPTER 04

Complete user journeys

Map the happy path, expected ambiguity, and no-camera scenario.

Journey A - destination is predownloaded

1. Before travel, the app recommends a city pack based on a searched destination. The traveller chooses text-only or full pack and sees download size.

2. Download completes; manifest signature, checksums, schema version and offline readiness are checked.

3. At the site, the traveller opens Scan; camera frame is embedded locally; GPS narrows candidate POIs only if permission was granted.

4. After scoring, show “Likely: Nalanda Mahavihara”, 1-3 alternatives, and source-backed overview. If no match clears policy, offer Search.

5. The user switches to “Architecture” and learns about a monastic plan, structures, materials and conservation, with citations.

6. If saved, the journal stores local note, visited time and chosen photograph without automatic upload.

Journey B - unknown or unsupported place

The photograph may show a modern shop, a generic courtyard, or a distant hill with no entry in the pack. TrueWay must say “This is not in the downloaded landmark set” rather than confidently guessing. It may offer approximate location results only when separately labelled as *nearby options*. The phone can still show the journal and map.

Journey C - photography prohibited

Select “Find by name / near me.” At Mahabodhi Temple, where Bihar Tourism lists cameras and mobile/electronic devices as prohibited, display an etiquette advisory before scanning. The traveller can view information before entering or where device use is permitted. [18]

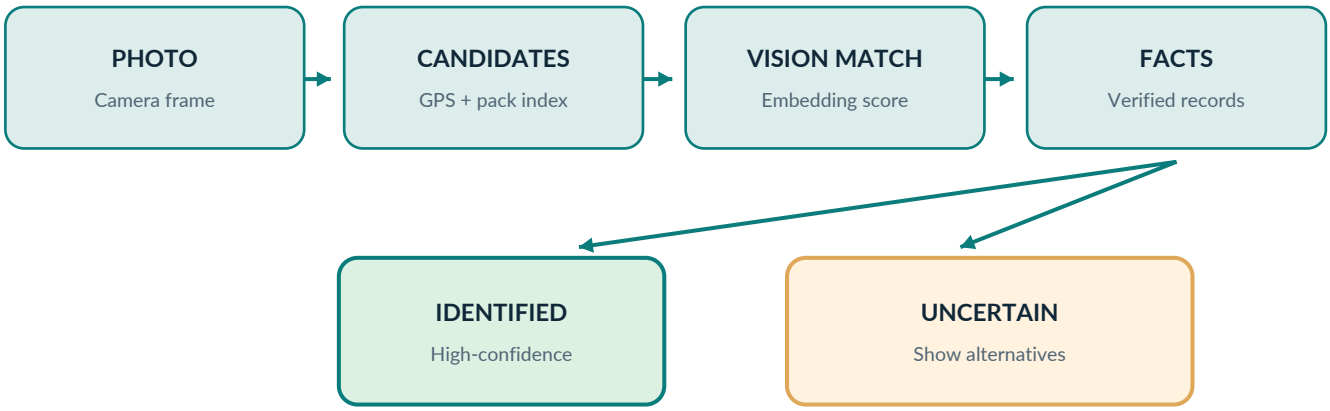
Journey D - no GPS / low battery

Offer a searchable list and saved route, allow reduced-power mode, and avoid continuously running the camera or generative model. A GPS fix may be slow or absent; never treat location as always available. Exit the scanner automatically when backgrounded.

CHAPTER 05

Landmark recognition: what will actually work

A visual search problem with controlled scope and calibrated uncertainty.



The system must support abstention, not just a best-guess label.

Figure 1. Recognition is a sequence of constrained candidate retrieval and verification, not unconstrained image-to-history generation.

Recommended baseline pipeline

- **1. Index:** curate 8-20 legal, varied reference photographs per supported landmark; embed them offline on a workstation; store vectors + landmark ID + image provenance.
- **2. Capture:** choose a sharp frame; normalize resolution/orientation and optionally detect blur.
- **3. Geo shortlist:** when available, use phone location, accuracy radius and region bounds as a probabilistic prior, not an absolute filter.
- **4. Visual retrieval:** compute an embedding and compare against reference vectors using cosine similarity or another validated metric.
- **5. Re-rank:** aggregate multiple image matches per landmark, optionally apply local-feature verification to reject lookalikes.
- **6. Decide:** use a threshold tuned on unknown-place photos; show top 3 and abstain below threshold.
- **7. Respond:** retrieve the verified place record and show source-backed facts with no fabricated certainty.

Model options and trade-offs

Candidate	Why consider it	Caution
Compact MobileNet/EfficientNet encoder	Small download; easy mobile quantization; can learn narrow corpus.	Training data quality can dominate model size.
EmbeddingGemma 2 Text-Vision	Official 388 MB option, shared text/image embedding capability. [1,2]	Recent model; benchmark memory, inference time, retrieval quality on target phones.
Gemma 4 E2B multimodal	Advanced generative reasoning, official 2,583 MB LiteRT-LM listing. [1,3]	Excessive as a mandatory MVP recognizer; RAM, heat and storage pressure.
Local feature matcher	Re-check building details after visual retrieval.	Less robust to occlusion and difficult views; adds complexity.

Training: cloud/workstation vs on-device

Recommended: fine-tune or train the image model **off-device** using a GPU workstation or cloud instance; export a quantized inference model to Android. “On-device AI” should mean inference runs locally. On-device fine-tuning or federated training is an advanced future option and brings battery, memory, privacy and model-poisoning complications.

Avoid accuracy traps

Use a location-disjoint and photographer-disjoint evaluation where possible. Landmark image collections contain repeated views, tourist duplicates and heavy class imbalance; random image-level splitting can leak nearly identical photos into train and test. GLDv2 is useful for research but its image licensing and long-tail categories require additional diligence. [4]

CHAPTER 06

System architecture and data flow

A small offline application with an optional distribution backend.

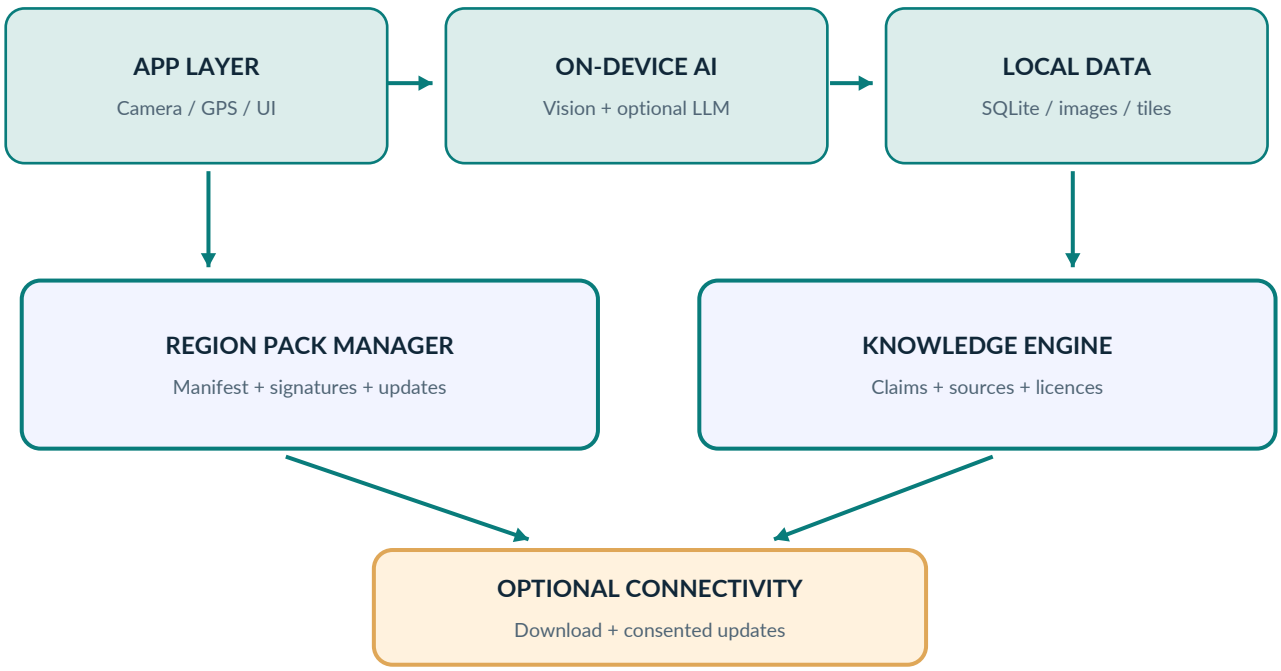


Figure 2. Runtime is entirely local after pack/model acquisition; connectivity is restricted to downloads, updates and explicit user actions.

Android runtime modules

Module	Responsibilities	Local artifacts
UI / accessibility	Scan, discover, history, sources, journal, pack management.	User preferences and offline state.
Capture / sensors	CameraX, GNSS location, photo quality control, permissions.	Temporary frame, coarse position.
Vision service	Run model, query index, calculate candidate scores.	Model weights, reference embedding index.
Knowledge service	Source-linked descriptions, claims, aliases, search.	SQLite tables and FTS index.
Map service	Display maps, POIs and simple trails where licensed.	Offline PMTiles/MBTiles or MapLibre region.
Pack service	Download / resume / verify / atomic activation / delete.	Manifest, chunk files, signatures.
Optional language engine	Grounded local Q&A;, translation, TTS/audio.	Model weights, context cache, language files.

Backend is not in the scanning loop

A Python / FastAPI publishing service can build and distribute packs, check editorial quality, and accept consented issue reports while users are online. Landmark identification and history reading must not call a server. A small CDN/object storage layer is enough for the initial pilot.

Data contracts and separation

Maintain three distinct IDs: **place_id** identifies a real-world place, **asset_id** identifies an image/audio source, and **claim_id** identifies a fact backed by one or more references. A visually plausible result is never a substitute for a place ID with known facts. Embedding vectors are indexes, not canonical truth.

Core tables

```
places(place_id, name, lat, lon, category, bounds, aliases)
claims(claim_id, place_id, claim_text, claim_type, date_start)
claim_sources(claim_id, source_id, pages_or_anchor, reviewed_at)
sources(source_id, title, url, publisher, licence, accessed)
assets(asset_id, place_id, type, creator, licence, source_url)
embeddings(asset_id, model_version, dims, vector_blob)
packs(pack_id, version, region_id, manifest_hash, status)
user_journal(entry_id, place_id, created_at, photo_path, note)
```

For SQLite full-text retrieval use FTS5; for nearby candidates, begin with a geohash/bounding-box shortlist and a validated distance calculation. An approximate-nearest-neighbour index is optional at the pilot scale; first measure brute-force latency. This avoids premature heavy dependencies.

CHAPTER 07

Offline content packs and storage economics

Pack architecture should be designed before adding global content.

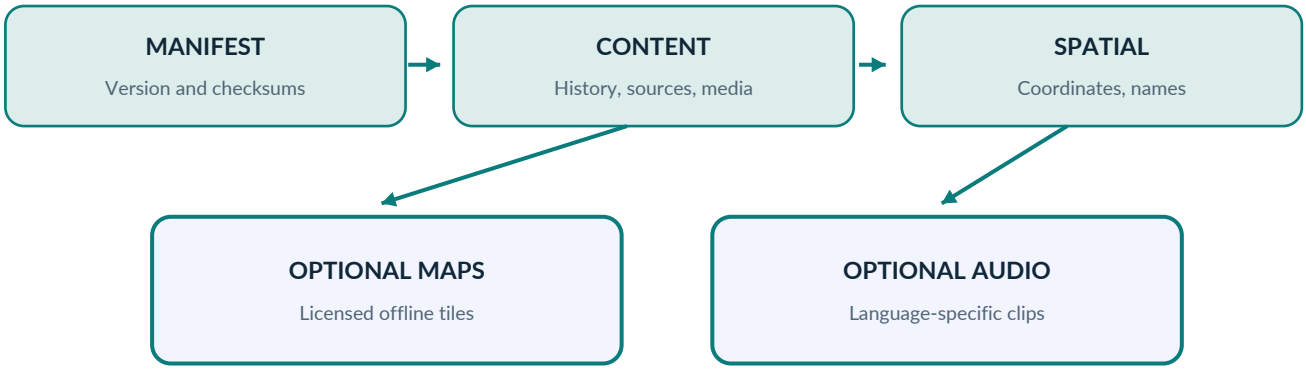


Figure 3. Text/history, geospatial content, audio and AI weights have different update cycles and should not be fused into one monolithic download.

Suggested pack types

Tier	Includes	Illustrative city-pack budget
Lite	Place IDs, names, geolocations, short claims, sources, mini thumbnail set.	5-25 MB of content (assumption).
Heritage	Full claims, reference images, multi-language text, selected thumbnails.	25-120 MB of content (assumption).
Media	Recorded narrations, richer imagery and detailed visitor material.	Additional 30-250 MB (assumption).
Maps	Licence-cleared vector map coverage with bounded zoom ranges.	Additional 25-200 MB (assumption).
Model	Vision model or optional LLM shared across regions.	Separate app-managed download: 388 MB / 2,583 MB examples. [1]

Important: illustrative pack sizes are engineering targets, not measured download footprints. Visual-reference density, locale count, tile zoom, recording bitrate and compression can change them substantially. For the first prototype, produce real test archives for 50, 100 and 500 landmarks and chart size growth.

Download state machine

Offered → Downloading → Paused / Resumable → Downloaded → Signature-verified → Unpacked → Indexed → Active. Update to a temporary version; never delete the known-good active pack until the new one passes validation. Roll back automatically when unpacking or index construction fails.

Manifest fields

```
{
  "pack_id": "in-bihar-buddhist-circuit",
  "pack_version": "2026.10.1",
  "schema_version": 2,
  "content_langs": ["en", "hi"],
  "place_count": 85,
  "components": ["facts", "assets", "vectors"],
  "files": [{"path": "places.db", "sha256": "...", "bytes": 0}],
  "min_app_version": "1.0.0",
  "licence_manifest": "ATTRIBUTIONS.json",
  "publisher_signature": "ed25519_signature_base64"
}
```

This is a suggested format; zero byte count and ellipsis are placeholders. Use canonical manifest serialization, strong checksums and public-key signature verification. Models need their own licence/version manifest.

When the user downloads a whole country

Partition by spatial regions, semantic circuits, or zoom/detail. Support country-level *index* first and optional city details later. Share a single model across all packs; de-duplicate images, audio and metadata by content hash where licensing permits. Use a storage budget and clean uninstall flow.

CHAPTER 08

Maps, navigation and positioning

Avoid making a map-service dependency a hidden online requirement.

Map foundations

MapLibre has official Android offline-region APIs, and recent MapLibre Native supports reading local PMTiles files. Map data has separate licence and sourcing constraints: OpenStreetMap's own raster and vector tile services do not permit bulk offline prefetched downloads. TrueWay should generate/contract permissibly licensed vector tile archives or use an explicit offline-permitted provider. [5,6,7]

Map functionality by maturity

Capability	MVP	Later
POI plotting	Show downloaded POI points on stored map or place list.	Styles, layers, indoor/museum maps.
Nearby search	Bounding-box + distance sort, GPS optional.	Context-aware recommendations.
Heritage trails	Curated paths with waypoints and length approximation.	Offline foot-routing graph and accessibility constraints.
Live directions	Not supported when offline; show disclaimer.	Dedicated routing engine/graph after field tests.
Hours and fees	Show last-verified timestamp and "may change."	Sync only with permitted authoritative feeds.

Precise GPS is not guaranteed

Phone GNSS can acquire position without internet, but assisted positioning speed, indoor signals, urban canyons and battery settings affect experience. Treat location as *evidence*, with accuracy radius and age; permit search and manual map-pick even when location is unavailable. Ask for location only while relevant to a user action.

CHAPTER 09

Knowledge engineering: facts before storytelling

Provenance is the feature that protects TrueWay from confident misinformation.

Content source hierarchy

- **Tier 1:** UNESCO World Heritage Centre, ASI / government heritage material, museums, peer-reviewed archaeology and original conservation documentation. [9,18,29,30]
- **Tier 2:** curated guidebooks, credible institutional publications, academic books with identifiable authors and dates.
- **Tier 3:** Wikidata structure and Wikipedia background used as leads; verify important contentious claims rather than accepting them automatically. [8]
- **Community contributions:** treat as unreviewed until editorial approval and recorded provenance.

Claims and uncertainty

Claim type	Sample presentation	Editorial rule
Documented	“The Mahabodhi site entered the World Heritage List in 2002.” [29]	Source + year + evidence link.
Historical interpretation	“The standing temple is generally dated to the 5th-6th centuries.” [29]	Attribution, range, and revisions retained.
Tradition / belief	“Buddhist tradition associates the site with enlightenment.” [29]	Attribute belief; do not assert a testable event as independently verified.
Contested	“Sources differ on the date/sequence...”	Describe competing evidence and avoid false precision.
Operational	“Entry rules as listed by Bihar Tourism, last checked...” [18]	Timestamp prominently; do not promise live status offline.

Suggested review workflow

Ingestion → normalise names/languages → de-duplicate places → map external IDs → extract claim candidates → verify against original context → attach source/licence → editorial sign-off → compile per-region SQLite pack → release tests → publish signed manifest. Claim revisions must be auditable, including reviewer and previous wording.

Example place record (illustrative)

```
place_id: in-bihar-bodhgaya-mahabodhi
name: Mahabodhi Temple Complex
aliases: Bodh Gaya Mahabodhi Temple
heritage_status: UNESCO World Heritage (2002)
summary: Gupta-period brick temple; major Buddhist pilgrimage site
source_ids: [UNESCO_WH_1056, BIHAR_TOURISM_MAHABODHI]
visitor_notice: device / camera restriction; check on-site rules
last_verified: 2026-10-10
uncertainty_note: chronology represents source interpretation
```

The Mahabodhi dating and inscription details above are independently corroborated by UNESCO. Visitor restrictions are from Bihar Tourism and may change, so they require refresh before release. [18,29]

CHAPTER 10

Optional local LLM and grounded conversations

Generative AI should explain evidence, not decide identity by improvisation.

The smallest working RAG loop

1. Choose a validated **place_id** and load only claims and paragraphs associated with that location.
2. Index local descriptions by SQLite FTS5; vector retrieval can be introduced once measured necessary.
3. For a question such as “Why was it built?” search the pack; pass 3-6 relevant passages with source IDs to a local model.
4. Constrain the model to the retrieved context and require a compact source list after factual statements.
5. When documents do not contain the answer, show “Not in the downloaded sources” and suggest a specific online update when connectivity returns.

Practical model-tier approach

Device tier	Recommendation	User-facing result
Low-end / limited storage	No local chat model; templates and prewritten summaries.	Fast, useful explanation with minimum footprint.
Mid-range	Compact quantized text model only after RAM/latency benchmarking.	Limited question-answering on selected phones.
Flagship	Evaluate Gemma 4 E2B with LiteRT-LM. [1,3]	More fluent sourced Q&A; optional multi-GB download.

Guardrail contract for the assistant

```
SYSTEM RULES FOR OFFLINE HERITAGE QA:
- Use ONLY supplied source passages for factual claims.
- Do not invent place names, years, citations or archaeological facts.
- Quote source IDs in the answer; mark myth/tradition as such.
- If evidence is missing, say so clearly.
- Do not give live opening times or prices without a freshness label.
- If image recognition was uncertain, do not present one match as fact.
```

Prompt instructions reduce risk but cannot guarantee correctness. Validate post-generation source IDs against actual retrieved passage IDs, check date strings and named entities where feasible, and fall back to extractive text when

validation fails. Latency, memory and battery measurements decide whether this feature ships.

Fine-tuning policy

Do not fine-tune a foundation LLM simply to memorise landmark facts. Keep factual content in regional packs so corrections do not force model retraining. Fine-tune small task-specific models offline only when pilot evaluations show measurable gain over simpler retrieval and carefully designed prompts.

CHAPTER 11

Localization, accessibility and inclusive UX

A traveller should not need a flagship phone or fluent English.

Languages

Begin with English + Hindi editorial content, then evaluate languages common among actual pilot visitors. ML Kit's Text Recognition v2 covers Devanagari, and its on-device translation supports 50+ languages after model download; Google cautions translation is designed for casual usage and accuracy varies. Serious cultural or historical claims should be human-reviewed in each language. [16,17]

Three levels of language quality

Level	Delivery	What gets reviewed
Human-authored source	Concise guide, facts, names, signboard terminology.	History, dates, religious and political sensitivity.
Human-reviewed translation	Hindi and pilot languages in pack.	Terminology, transliterations and accessibility.
Machine-generated optional	Offline on-device translation with a disclaimer.	Label origin; avoid asserting perfect equivalence.

Accessibility & ethical design

- Large text, high contrast, simple language, adjustable playback speed, subtitles and offline text-only mode.
- Provide a visual "camera restricted" advisory before capture at flagged sites; do not encourage unsafe, disrespectful, or prohibited behaviour.
- Offer audio narration created from reviewed scripts; offline TTS availability differs by OS/device/voice and must be tested.
- Do not show exact location history by default; journal location metadata should be user-controllable.
- For reconstructions, label illustrative or hypothetical architectural elements separately from observed evidence.

Suggested home navigation

Explore (nearby + search), **Scan** (camera and alternatives), **Saved** (journal and lists), and **Downloads** (regions, languages and models). Historical sources should be one tap from the summary, not hidden behind a paid plan.

CHAPTER 12

Licensing, content rights and site permissions

The data pipeline must preserve legal obligations at item level.

Source-specific compliance matrix

Input source	Permission / caveat	Required operational handling
Wikidata	Structured statements under CC0. [8]	Record source URL and source confidence; attribution good practice.
Wikimedia Commons	Media may be public domain or CC with varying attribution/share-alike obligations. [10,25]	Preserve asset author, file page, licence, changes and attribution display.
Wikipedia articles	Text subject to project-specific Creative Commons terms.	Record revision ID, author/attribution route; counsel reviews adaptation and SA obligations.
UNESCO World Heritage dataset	Listed as CC BY-SA 4.0 on data portal. [9]	Attribute UNESCO and examine derivative/share-alike requirements.
UNESCO property descriptions	Property pages may have CC BY-SA IGO licences. [29,30]	Check item-specific notice; prefer summarised original text with source link.
OpenStreetMap data	ODbL with attribution and possible share-alike triggers. [7]	Map notices; document derived database separation and export policy.
OSMF tile servers	No prefetch/offline bulk download. [6]	Use independent lawful tile generation or offline-licensed provider.
Google Landmarks v2	Image labels/metadata are research inputs; image rights differ. [4]	Do not assume benchmark image URLs grant commercial reuse.
Protected heritage sites	Different rules for photography, filming and device use. [18,21]	Location manual mode; obtain site-specific permissions for commercial production.

Legal/asset manifest

Every media item needs: canonical source URL, creator, licence identifier/version, attribution string, allowed transformations, retrieval date, modification log, withdrawal/contact route and distribution flags. Block pack publication when rights are unknown, disputed, or incompatible with intended commercial use.

ASI filming rules and permission processes should be checked separately from ordinary visitor snapshots; a commercial field-data campaign is not automatically covered by tourist-photography expectations. [21]

Product name and trademark

Treat “TrueWay” as a working product name. Run IP India trademark class searches, app-store conflict review, domain checks and counsel review before spending on brand campaigns. A web-result name check is not legal clearance.

CHAPTER 13

Privacy, security and compliance

Offline processing narrows the data surface but does not eliminate obligations.

Privacy defaults

- **Photos:** inference happens in memory; do not upload or retain the scan unless the user saves it.
- **Location:** request while scanning / nearby; store exact history only with explicit journal action.
- **Analytics:** offer opt-in telemetry, aggregated non-identifying signals, and easy revocation.
- **Permissions:** never require continuous background location for core functionality.
- **Deletion:** provide one-tap clearing for photos, journal, search history and downloaded models/packs.
- **Children:** avoid introducing child-focused profiling and targeted advertising without specialist compliance review.

Indian compliance context

India's Digital Personal Data Protection Rules, 2025 and their enforcement timeline were formally published in November 2025; obligations and commencement dates are phased. The team should verify the applicable stage at launch, publish a plain-language privacy notice, establish grievance handling, and get counsel review before collecting personal data. This document is not legal advice. [20]

Security threat model

Threat	Impact	Mitigation
Tampered pack	Malicious content, false facts or code/data exploit.	Signed manifest, SHA-256 per file, parser hardening and rollback.
Unsafe model bundle	Unexpected behavior or supply-chain issue.	Pin hashes, licence, provenance, supported devices, model version.
Location leakage	Exposure of sensitive visits and habits.	Default no-upload, local journal, minimised logs, access controls.
Misleading fact / hallucination	Cultural harm and eroded trust.	Claim-level sourcing, abstention, editorial corrections and clear reporting.
Large pack exhaustion	Storage or device instability.	Quota check, compressed streaming, cancellation and cleanup.
Analytics reidentification	Privacy / regulatory risk.	Opt-in, data minimisation and retention limits.

CHAPTER 14

Research and evaluation methodology

Build a repeatable benchmark before claiming the system recognizes landmarks.

Pilot dataset design

Proposed baseline: 50-100 supported POIs, with approximately 10-30 reference photos per POI across viewpoint, weather, occlusion and distance. A 100-POI / 20-image reference corpus has 2,000 reference photographs. These are planning numbers; collect only lawfully reusable or permitted imagery. Include a separate test set of unseen photographers and visits.

Unknown and edge-case testing

Collect unknown scenes from the same geographical corridor: non-landmark buildings, adjacent courtyards, interiors, road signs, lookalike temples, replicas, blurry images and tourist crowds. Without this negative set, a recognizer can show impressive closed-set accuracy while mislabelling almost every real-world unknown.

Metric definitions and release gates

Metric	Meaning	Proposed validation target
Top-1 accuracy	Correct first match on known-set held-out images.	At least 85% on pilot set; no claim before measurement.
Top-3 recall	Correct landmark among three suggestions.	At least 95% known-set target.
False accept rate	Unknown images assigned a “known” identity.	At most 5% at launch threshold, ideally lower.
Coverage at accepted confidence	Fraction of known examples passing threshold.	Report alongside accuracy; never hide abstentions.
End-to-end latency	Photo capture to identified card.	P50 under 2.5 s; P95 under 6 s on nominated baseline phone.
Battery / heat	Cost per scans and 15-min guided walk.	Thermal / battery limit decided from device field trial.
Claim traceability	Factual display tied to local references.	100% for curated historical statements.
Airplane-mode pass rate	Successful flows without any internet.	100% of documented core flows after installation.

Evaluation splits and fairness

Split by photographer, visit/session and location condition; watch for similar images and duplicates. Measure day/night, crowded/empty views, partial occlusion, low-end phones, linguistic groups, and rare/repeated landmark categories separately. A model good at Mahabodhi may still fail on repetitive Nalanda bricks.

Human validation protocol

Have at least two local heritage reviewers independently examine a representative sample of claims and image labels. Adjudicate disagreements, log edits, and reserve “unknown” cases. Quantify factual error rate, not only model retrieval accuracy.

CHAPTER 15

Implementation and delivery plan

A realistic 24-week milestone plan for a serious pilot.

Weeks 1-3	Product + dataset
4-7	Vision baseline
8-11	Offline experience
12-15	Trust + localization
16-20	Field pilot
21-24	Hardening

Figure 4. Suggested sequence; tasks can overlap, but every release requires a stable offline regression test.

Phase	Deliverables	Exit gate
Weeks 1-3: Discovery	UX interviews (15-25), rights registry, taxonomy, 50-100 landmark shortlist.	Clear target devices and fully legal starter dataset.
Weeks 4-7: Retrieval	Reference photos, embedding pipeline, local ANN/brute-force test, threshold study.	Reproducible baseline with known/unknown metrics.
Weeks 8-11: MVP	Kotlin UI, camera, SQLite facts, region pack installation, manual search.	No-network end-to-end task completion.
Weeks 12-15: Trust	Source attribution, corrections, safe fallback, Hindi, privacy and audio text.	Expert content sign-off and security review.
Weeks 16-20: Field	Bodh Gaya/Nalanda/Rajgir permitted field tests, QA, telemetry with consent.	User outcome + latency + coverage metrics documented.
Weeks 21-24: Ship candidate	Optimize models, polish downloads, crash fixes, onboarding and Play policies.	Pilot release checklist passed.

Team structure

Role	Suggested allocation	Responsibilities
Founder / product owner	0.5-1 FTE	Product decisions, partnerships, content priorities.
Android engineer	1 FTE	App, CameraX, storage, maps and CI.
ML / retrieval engineer	1 FTE	Model benchmark, inference integration, evaluation.
Heritage researcher / editor	0.5-1 FTE	Claims, sources, licences, localization QA.
Designer / UX researcher	Part time	Usability, accessibility, testing scripts.
QA / field tester	Part time	Device matrix, airplane-mode and field validation.

Engineering process

Use a monorepo or cleanly versioned repos for Android app, dataset pipeline and pack builder. Every build should run unit tests, schema migration tests, offline “zero network” tests, model load smoke tests, licence audit and at least one physical-device benchmark.

CHAPTER 16

Budget and operating economics

All monetary values below are illustrative Indian project-planning assumptions, not vendor bids.

24-week staffed pilot - bottom-up budget

Cost item	Assumption	Amount
Android development	6 months × INR 100,000	INR 6.00 lakh
ML / retrieval engineering	6 months × INR 120,000	INR 7.20 lakh
Heritage research / editor	5 months × INR 60,000	INR 3.00 lakh
UI/UX and user research	2 months × INR 70,000	INR 1.40 lakh
Field/QA support	3 months × INR 50,000	INR 1.50 lakh
Compute, devices and software	Pilot allowance	INR 1.00 lakh
Travel, permits, content prep	Pilot allowance	INR 1.50 lakh
Legal/IP/licensing review	Pilot allowance	INR 0.80 lakh
Subtotal	Excludes tax / founders not otherwise listed	INR 22.40 lakh
Contingency	15% of subtotal	INR 3.36 lakh
TOTAL	**Illustrative staffed project**	**INR 25.76 lakh**

Interpretation: team monthly rates are model inputs, not surveyed compensation benchmarks. Founder-developed pilots may spend much less cash by using existing skills and free/open tools, but donated time is still an economic cost. Commercial agencies may quote materially more. Seek three comparable local quotations once scope is frozen.

Three budget levels

Approach	Indicative cash outlay	Trade-off
Founder/student feasibility	INR 1.5-5 lakh + unpaid effort	Prototype, 15-30 places, limited editorial review; not a production guarantee.
Lean staffed field pilot	INR 15-30 lakh	Focused 50-100 POI region; production-minded engineering and QA.
Multi-city product rollout	INR 40 lakh and above	Content expansion, multilingual experts, richer media and higher QA burden.

Pack distribution sensitivity (illustrative)

Model bandwidth as **downloads × average bytes per download × CDN egress rate**. Example: 10,000 city-pack downloads at 80 MB each create about 800 GB of data transfer. At an *assumed* INR 12/GB, that is ~INR 9,600 of egress before taxes, requests, storage and cache overhead. A 2.58 GB optional LLM downloaded 10,000 times would instead generate ~25.8 TB. Do not make large-model downloads the default.

Where budgets are most sensitive

Content rights clearance, new language quality, audio production, device coverage and field iteration could exceed generic cloud costs. Historical research is a production dependency, not a one-time data scrape. The business model should price the cost of maintaining and reviewing the knowledge base.

CHAPTER 17

Business model, pricing experiments and growth

Test willingness to pay rather than infer it from tourism visit totals.

Four monetization hypotheses

Model	Hypothesis to test	Main risk
Freemium consumer	Free core scan/search; richer guided circuits or audio as paid add-ons.	People may view cultural knowledge as a free utility.
Premium annual	Offline itinerary, multilingual narration, large pack allowance.	Travel is intermittent; subscription resistance.
B2B heritage partners	Custom editorial packs, branded delivery, analytics only with lawful consent.	Long procurement and content approval cycles.
Institutional / education	Licences for curated architectural-history datasets and trails.	Requires faculty/heritage validation and specialized workflows.

Pricing ideas - experiments only

Consider testing one-off circuit packs around **INR 99-299** and an annual premium option around **INR 799-1,499**. These are survey/test ranges, *not* researched market-clearing prices. Essential source attribution, low-vision support and basic manual search should stay accessible. Evaluate willingness to pay with real users before integrating billing.

Example conversion economics



Figure 5. Qualitative adoption funnel only; band widths do not represent measured conversion data.

Measure activation as: a new install completes a destination pack, identifies or manually finds one supported place, and reads a history view without network. Retention should mean a return visit on a separate travel day, not merely an app open.

Pilot interview questions

- What did you do last time you wanted historical details at a monument?
- Did you have network access? Would you prepare a pack ahead of time?
- Would you trust a guessed monument name? What would convince you it was right?
- Which is more valuable: quick identification, audio narration, or offline route planning?
- Would you pay once for a curated circuit? What competing free tool would you otherwise use?

CHAPTER 18

Bihar heritage pilot: site and content program

A concrete starting geography that balances importance with reasonable scope.

Why this circuit

Bihar's official tourism material identifies Bodh Gaya, Rajgir and Nalanda among its key Buddhist destinations, and UNESCO lists the Mahabodhi Temple Complex and Archaeological Site of Nalanda Mahavihara as World Heritage properties. This produces two high-value research anchors and multiple additional points of interest for testing. [19,29,30]

Pilot coverage plan (proposal, not a current dataset)

Cluster	Candidate themes and categories	Initial target
Bodh Gaya	Temples and monasteries, memorials, contextual architecture, permitted street viewpoints.	20-30 POIs
Rajgir	Hillside heritage, routes, religious and archaeological sites, viewpoints.	15-25 POIs
Nalanda	Mahavihara structures, archaeological plan, museums and interpretive viewpoints.	15-25 POIs
Buffer/negative scenes	Everyday shops, streets, generic shrines, incomplete views, lookalikes.	Not POIs; separate negatives

Editorial sample: Mahabodhi Temple Complex

Verified content: UNESCO notes the initial temple built under Emperor Asoka in the 3rd century BCE and the extant brick temple mainly dating to the 5th-6th centuries CE; it was inscribed on the World Heritage List in 2002. [29]

Suggested app narration: "You are exploring one of the most important sites associated with Buddhist enlightenment. The present brick temple reflects architectural work from the Gupta period. You can explore its historical phases, symbolic forms and the heritage conservation story." This is original illustrative copy, not a quotation from UNESCO.

On-site guardrail: Bihar Tourism lists mobile, camera and electronic device restrictions; the app must support pre-visit reading and post-visit exploration, not prompt visitors to violate site rules. [18]

Editorial sample: Nalanda Mahavihara

UNESCO describes a monastic and scholarly complex with viharas, shrines, stupas and material remains spanning long historical periods. Architectural interpretive topics can include axial planning, brick masonry, courtyards, monasteries and conservation methods, while drawing a line between surviving fabric and reconstructed hypotheses. [30]

Local research collaboration

Seek voluntary review from knowledgeable regional historians, archaeology researchers, language editors, and tourism stakeholders. Obtain image permissions independently. Interview guides and pilgrims as design participants, but obtain explicit consent and avoid treating one community perspective as a universal historical account.

CHAPTER 19

Risks, failure modes and mitigations

The leading risks are accuracy, content rights and field adoption - not merely model coding.

Risk / severity	Likely symptom	Mitigation / release block
False identification / Critical	Generic temple confidently mislabelled.	Open-set threshold; alternatives; manual mode; severe false-accept release gate.
Hallucinated history / Critical	Invented dynasty, builder or date.	Retrieval-only claim path; approved scripts; cite/verify claims.
Image licence issue / High	Unlicensed photos distributed in city pack.	Asset-rights register; automated licence gate; remove or replace.
Unusable download / High	Pack too large, partially corrupt, or lost during update.	Chunk resume, signed manifest, atomic activation, quota check.
Device exclusion / High	Low-end Android cannot load AI model.	Tiny baseline, device tiers, no mandatory local LLM.
No-camera heritage rule / High	Visitor blocked at entrance.	Manual lookup and etiquette flag; avoid incentivizing capture.
Insufficient coverage / High	Travellers scan unsupported places repeatedly.	Transparent region counts; coverage map; add based on opt-in requests.
Stale operational facts / Medium	Old fees/hours presented as current.	Dated snapshot and "verify locally" warning.
Weak monetization / High	Users love demo but do not pay.	Early payment experiments; B2B pilots; strong free value.
Platform change / Medium	Inference SDK / API breaks.	Pin stable versions, compatibility layer, regular device CI.

Stop/go decisions

Stop: if more than 5% of unknown scenes are incorrectly accepted at the chosen threshold, if historical statements lack source traceability, or if the app fails airplane-mode tests. **Re-scope:** if model weight / battery targets cannot run on the lowest targeted phones. **Go:** after measurable user success and a workable rights framework are demonstrated.

CHAPTER 20

Acceptance checklist and test plan

A launch is a set of passed tests, not a date on a roadmap.

Critical functional tests

Case	Setup	Pass criterion
OFF-001	Airplane mode; pack fully installed.	Photo scan shows candidate or honest abstention without network.
OFF-002	Remove all network permissions after install.	History, search, journal and map still work.
OFF-003	Interrupt pack download midway.	Resumes without corruption or duplicate assets.
OFF-004	Corrupt a file in a staged update.	Checksum fails and prior pack remains active.
ML-005	Generic building with landmark-like arches.	No confident historic identity without support.
ML-006	Two nearby visually similar ruins.	Candidates shown; no arbitrary false certainty.
UX-007	Select photography-prohibited POI.	Manual search route and advisory offered.
TRUST-008	Remove one factual source from pack.	Claim no longer surfaces unsupported as verified.
PRIV-009	Decline location access.	Search + scan continue; no permission loop.
DEV-010	Run lowest spec supported phone 15 minutes.	No crash / unacceptable thermal or memory condition.

Physical-device matrix

At least one entry device, one mid-range device, and one flagship Android; include varied OS versions and chipsets. Test with low available storage, 10% battery, screen-reader mode, poor GPS, and denied permissions. Do not list a device as supported solely because its OS API level is high enough.

Field UX scorecard

KPI	What it tells you	Proposed target
First valuable result	Time from first open to first useful history screen.	Within 3 minutes after predownload.
Task completion	User reaches history without outside assistance.	At least 80% during moderated pilot.
Correctness perception	User knows whether answer is sourced or uncertain.	At least 85% comprehension in test questions.
Return intent	Would traveller use it on the next trip?	Measure; no invented target.
Content satisfaction	Was history understandable, accurate and interesting?	Report median and reason themes.

CHAPTER 21

Concrete starter backlog and decision questions

Turn the research into repository milestones.

First 14 days

- **Days 1-2:** create Android Kotlin project, app navigation, CameraX proof of concept and a mock landmark card.
- **Days 3-4:** define place/source/asset schema and write 10 editorially sourced landmark records.
- **Days 5-6:** obtain 10-20 legal reference images for each of 15 demonstrator landmarks; document rights.
- **Days 7-8:** run initial embedding baseline offline on workstation; calculate top-1, top-3 and unknown rejection.
- **Days 9-10:** export model/index to one Android test phone; inspect memory, latency and cold start.
- **Days 11-12:** implement zipped pack + manifest import and explicit airplane-mode acceptance test.
- **Days 13-14:** user test with 5 people; document issues; select the most viable model/pilot scope.

Priority epics and sample acceptance stories

Epic	User story	Done when
SCAN	As a traveller I photograph a supported monument.	Top candidates appear offline; abstention enforced.
KNOW	As a visitor I can see why a place matters.	Curated text + sources open from card.
PACK	As a traveller I download only my destination.	Install, resume, update, quota and delete work.
SEARCH	As a visitor I cannot use my camera here.	Name / nearby discovery works with no image.
TRUST	As a reader I want to know the source.	Attribution and last-reviewed date show in context.
PRIVACY	As a user I do not want my travel photo uploaded.	Processing local by default and one-tap local deletion.

Unresolved decisions requiring experiments

- Which Android devices and minimum free storage should TrueWay target?
- How frequently do travellers actually scan a landmark, and when is manual search preferable?

- How many legal, varied reference photos are available per pilot POI?
- What is the acceptable false-identification risk for a public launch?
- Can a compact model beat the 388 MB embedding option in accuracy per watt and storage?
- Will travellers pay for audio, language packs, guided trails or partner-curated packages?

> **Primary recommendation**
>
> Build the **first 15-place airplane-mode demonstration** before licensing a global image corpus or integrating a large language model. If users find accurate results compelling, expand the corpus and test the 50-100-place Bihar pilot. That sequence protects both development time and factual trust.

CHAPTER 22

Appendix A - technical pseudocode

An implementation sketch, not production-ready code.

Inference and abstention

```
fun identify(photo, location, activePack): RecognitionResult {
  val vector = encoder.embed(photo)
  val matches = index.topK(vector, k = 20)
  val ranked = geoRerank(matches, location, activePack)
  val top = ranked.take(3)
  val certainty = calibrator.score(top, photo.quality)
  if (top.isEmpty() || certainty < threshold) {
    return Unknown(topAlternatives = top)
  }
  val place = db.lookup(top.first().placeId)
  return Identified(place, alternatives = top, score = certainty)
}
```

The threshold must be fitted on held-out known and unknown photos, not chosen as an arbitrary similarity value. Return candidate scores as an internal diagnostic until calibration is validated. Do not display “98% confidence” without measuring probability calibration.

Sourced response policy

```
fun explain(placeId, userQuestion): Answer {
  val evidence = localIndex.retrieve(placeId, userQuestion)
  if (evidence.isEmpty()) return UnsupportedQuestion()
  if (!device.canRunOptionalLlm()) return extractiveAnswer(evidence)
  val draft = localLlm.generate(evidence, userQuestion)
  return if (citationsValidate(draft, evidence)) draft
         else extractiveAnswer(evidence)
}
```

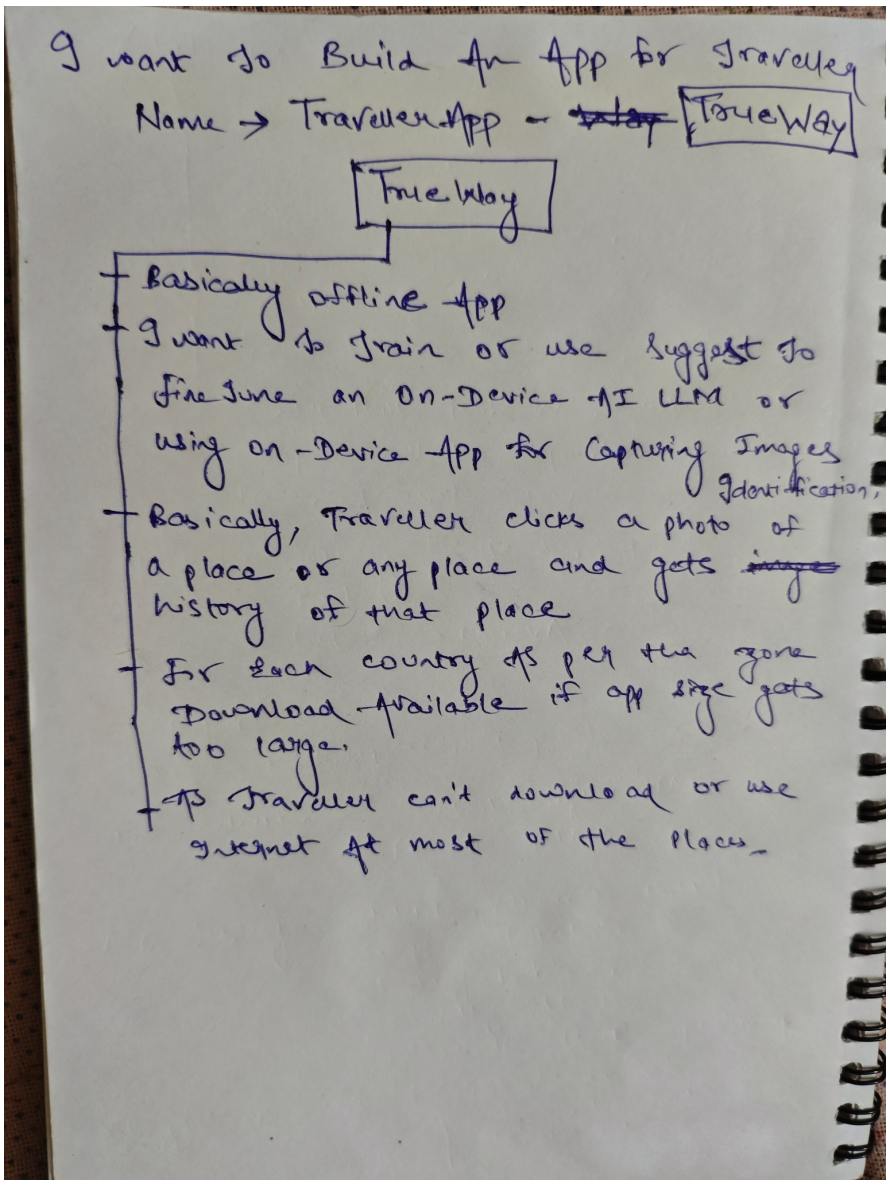
Minimum local indices

A SQLite B-tree keyed by place ID; FTS5 on place names, aliases and descriptions; simple geographic bounds table; and a binary embedding matrix or a validated vector-search extension. An HNSW/ANN library becomes useful only after baseline scan time exceeds user targets.

CHAPTER 23

Appendix B - originating handwritten concept

Interpreted from the founder-provided note; retained to document original intent.



Original concept note supplied with the brief. It calls for an offline-first app, on-device AI image identification, regional downloads, and a historical explanation after a photo is captured.

Research translation

The handwritten idea says, in essence: build an offline traveller app called TrueWay; use or fine-tune an on-device AI model for photographing and identifying places; show their histories; and distribute downloads by country/region so storage does not become too large. The detailed architecture in this report makes those objectives technically testable.

CHAPTER 24

Sources, verification notes and limitations

Primary sources prioritized; all links reviewed or retrieved 10 October 2026.

Bracketed numbers [1] ... [30] refer to sources below. A source documents a fact or platform claim; it does not validate TrueWay model accuracy, user adoption or its estimated budget. For evolving software and law, verify the live version before implementation.

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Methodology and interpretation

Sources were chosen to prioritize official SDK docs, vendor model cards, government tourism/material, UNESCO heritage descriptions and original licence policies. Market indicators describe tourism activity, not conversion. Financial planning figures and engineering targets are explicitly hypothetical; device support and recognition performance require hands-on benchmarking. This report does not certify trademark availability, data licensing compatibility for any planned derivative database, or compliance with future regulation.